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TEST REPORT

EXPANSION AND BLEEDING OF FRESHLY MIXED GROUT IN THE LABORATORY / SITE

Test Standard: ASTM C940-22 | In-house Method: STP04-GTT-C940 rev 2

Report No.		**Job Ref. No.**	
Client		**Date of Issue**	
Project			
Site Location		**Contract / W.O. No.**	
Batch / Sample ID & Pile No.		**Date Tested**	
Test Procedure	STP04-GTT-C940 rev 2 (Eff. 17 Apr 2026)	**Ambient Temp. (23 ± 2 °C)**	

1. Grout Mix Information & Calibrated Apparatus (Placed on Vibration-Free Level Surface ≤ 3 min)

Cement / Admixture / W/C		**1000 mL Cylinders (×3)**	LT-GMC / PMC-_____
Fresh Grout Temp. (°C)		**25 mL Graduate & Therm.**	LT-_____ / LT-TM-___
Initial Vol H_1 (800±10 mL / 100±5 mm)	Spec 1: ___ | Spec 2: ___ | Spec 3: ___	**Unit Mode & Spirit Level**	☐ mL ☐ mm | LT-LV-01

2. Time-Series Readings for 3 Replicate Specimens (15m / 30m / 45m / 60m / 2h / 3h / 24h)

Elapsed Time	Specimen No. 1			Specimen No. 2			Specimen No. 3		
	Water H_2	Grout H_3	Bleed %	Water H_2	Grout H_3	Bleed %	Water H_2	Grout H_3	Bleed %
0 min (H1)									
15 min									
30 min									
45 min									
60 min									
2 hour									
3 hour									
24 hour									

3. Final Calculated Results Summary — Reported to Nearest 0.2% (STP04 Cl. 6.1–6.4 & Cl. 8.2–8.5)

Calculated Property (Nearest 0.2%)	Specimen 1	Specimen 2	Specimen 3	Mean (0.2%)
Grout Expansion % = $(H_3 - H_1) / H_1 \times 100$				
Prescribed-Interval Bleeding % = $(H_2 - H_3) / H_1 \times 100$				
Free (Combined) Expansion % = $(H_2 - H_1) / H_1 \times 100$				
Decanted Bleed Water H_w & Final Bleeding % = $H_w / H_1 \times 100$				

Measurement Uncertainty & Remarks: Evaluated per MU-GTT-C940-1 (2026-04-21): Expanded measurement uncertainty U95 = ± 0.2% for bleeding and ± 0.2% for expansion (coverage factor k = 2, ~95% confidence). All values rounded to nearest 0.2%.

Tested & Recorded By (Qualified Operator):

Name & Signature Date: _____

Checked & Approved By (HOKLAS Approved Signatory):

Authorized Panel: Lam Hoi Fan, James (JL) / Tang Man Fu (MFT)

Authorized Signature & Company Chop Date: _____

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